

Decision and Risk Analysis

Lecture 2: **Bayes' Rule**

DAEN 427/ISEN 427/ISEN 627

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Let's suppose that you are an air traveler along with 99 other passengers from your flight, each of whom, like you, checked one item of luggage.

A recording piped through the speakers reminds you that: *"Many bags look alike. Please check your bag carefully before exiting the terminal."*

Indeed, your bag is one of the most popular models on the market, a medium-sized black rectangular case **used by 5% of all travelers**.

Given your distance from the luggage chute, you can discern only the shape, size, and color of the bags that emerge, not any detailed markings on the bags.

- a. Let's suppose that the **first bag** from your flight to enter the luggage carousel indeed has the same shape, size, and color as yours. Is it yours? Explain (i.e., calculate the posterior probability that the 1st bag is yours).
- b. Now suppose that we continue to wait for our bag to appear, failing to see it among the first 98 bags to enter the carousel (i.e., only 2 bags are left to come off the airplane). Calculate the posterior probability that the **99th** bag, which also matches ours in shape, size, and color, is our own.

We're Using R Instead of Python

Python

- “Can do everything” ... installs n packages, breaks after update
- Stack Overflow becomes your TA

R

- Literally for statistics; `lm(y ~ x)` and move on with your life
- Excellent, open-source, keeps us focused on data analysis, ... and I enjoy sleeping at night!

What I Need From You: learn concepts, analyze data, pass the class!

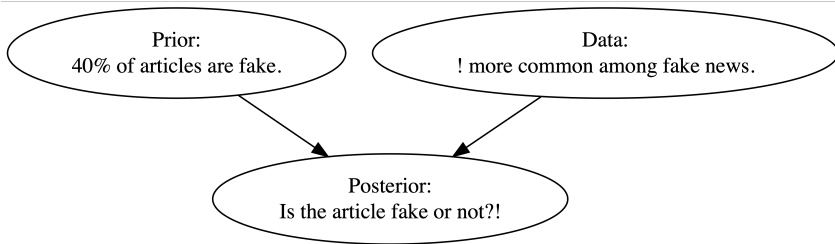
What I Do Not Need: a 2-hour debate about virtual environments.

Dataset structure

```
1  # Load packages
2  library(bayesrules)
3  library(tidyverse)
4  library(janitor)
5
6  # Import article data
7  data(fake_news)
8
9  fake_news %>%
10     tabyl(type) %>%
11     adorn_totals("row")
12
13 #   type    n percent
14 #   fake   60     0.4
15 #   real   90     0.6
16 # Total 150     1.0
```

- 150 news articles: 60 fake, 90 real
- $P(\text{Fake}) = \frac{60}{150} = 0.40$. Before looking at the headline, the prior probability that an article is fake is 40%.

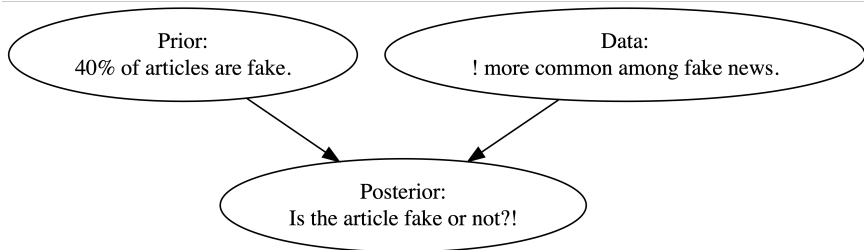
Bayesian reasoning to estimate whether a news article is fake:



Observed headline pattern:

```
1 # Tabulate exclamation usage and article type
2 fake_news %>%
3   tabyl(title_has_excl, type) %>%
4   adorn_totals("row")
5 # title_has_excl fake real
6 #           FALSE  44   88
7 #           TRUE   16    2
8 #           Total  60   90
```

Exclamation Points as Evidence



Title Has "!"	Fake	Real	Total
Yes	16	2	18
No	44	88	132
Total	60	90	150

Exclamation Points as Evidence

Title Has "!"	Fake	Real	Total
Yes	16	2	18
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$$P(! \mid \text{Fake}) = \frac{16}{60} \approx 0.267, P(! \mid \text{Real}) = \frac{2}{90} \approx 0.022$$

An exclamation point increases the probability that an article is fake.

$$P(\text{Fake} \mid !) = \frac{P(\text{Fake}) P(! \mid \text{Fake})}{P(!)}$$

Building a Bayesian Model for Events

Prior Probability Model

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-

Conditional Probability & Likelihood

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Conditional vs Unconditional Probability

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-
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Independent Events

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Probability vs Likelihood

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Normalizing Constants

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Calculating Joint and Conditional Probabilities

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Posterior Probability Model via Bayes' Rule

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Posterior Simulation

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Building a Bayesian Model for Random Variables

Prior Probability Model

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Discrete Probability Model

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The Binomial Data Model

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Conditional Probability Model of Data Y

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The Binomial Model

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The Binomial Likelihood Function

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Probability Mass Functions vs Likelihood Functions

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Normalizing Constant

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Posterior Probability Model

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Bayes' Rule for Variables

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-

Posterior Shortcut

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Proportionality

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-
-

Posterior Simulation

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-
-

Quiz 1

A medical test correctly detects a disease 95% of the time when the disease is present. This value represents:

- a. Prior probability
- b. Posterior probability
- c. False positive rate
- d. Likelihood

Quiz 2

A university uses an AI system to detect plagiarism.

- 5% of assignments are actually plagiarized.
- The AI flags plagiarized assignments 92% of the time.
- The AI incorrectly flags honest assignments 8% of the time.

If an assignment is flagged, what is the probability it was actually plagiarized? Briefly discuss whether the AI system seems reliable.

Quiz 2

Answer:

$$P(P \mid F) = \frac{0.92 \cdot 0.05}{0.92 \cdot 0.05 + 0.08 \cdot 0.95} \approx 0.377$$

Approximately 37.7% chance that it is actually plagiarism.
The system catches most plagiarism cases, but because plagiarism is rare, many flagged assignments are false positives.

a. Let's suppose that the first bag from your flight to enter the luggage carousel indeed has the same shape, size, and color as yours. Is it yours? Explain (i.e., calculate the posterior probability that the 1st bag is yours).

Answer: Let Y = first bag is yours, and M = first bag matches.

$$P(Y) = \frac{1}{100}$$

$$\mathcal{L}(Y^c | M) = P(M | Y^c) = 0.05$$

$$\mathcal{L}(Y | M) = P(M | Y) = 1$$

$$\begin{aligned} P(Y | M) &= \frac{P(M | Y)P(Y)}{P(M)} = \frac{P(M | Y)P(Y)}{P(M | Y)P(Y) + P(M | Y^c)P(Y^c)} = \\ &= \frac{1 \cdot 0.01}{1 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.01}{0.0595} \approx 0.168 \end{aligned}$$

So even though the first bag matches, it is probably not yours.

b. Now suppose that we continue to wait for our bag to appear, failing to see it among the first 98 bags to enter the carousel (i.e., only 2 bags are left to come off the airplane). Calculate the posterior probability that the 99th bag, which also matches ours in shape, size, and color, is our own.

Answer:

$$P(Y | M) = \frac{1 \cdot \frac{1}{\text{total of bags remaining}}}{1 \cdot \frac{1}{\text{total of bags remaining}} + 0.05 \cdot \frac{\text{total of bags remaining} - 1}{\text{total of bags remaining}}}$$

(if 2 bags remain)

$$= \frac{1 \cdot \frac{1}{2}}{1 \cdot \frac{1}{2} + 0.05 \cdot \frac{1}{2}} = \frac{1}{1 + 0.05} \approx 0.9524$$

There is a 95% chance that the 99th bag is yours, given that it looks like yours.

Or, it is **19 times more likely** to be yours than not yours ($0.95/0.05=19$).

Homework

- Submit your answers before the next class.
- You may discuss with peers but submissions must be individual.
- Responses may be shared anonymously during class discussion!

Homework

Question 1

Define the following events for a resident of a fictional town as:

- A = drives 10 miles per hour above the speed limit,
- B = gets a speeding ticket,
- C = took statistics at the local college,
- D = has used R,
- E = likes the music of Prince, and
- F = is a Texan.

Several facts about these events are listed next.

Homework

Question 1, cont.

Specify each of these facts using probability notation, paying special attention to whether it's a marginal, conditional, or joint probability.

- 73% of people that drive 10 miles per hour above the speed limit get a speeding ticket.
- 20% of residents drive 10 miles per hour above the speed limit.
- 15% of residents have used R.
- 91% of statistics students at the local college have used R.
- 38% of residents are Texans that like the music of Prince.
- 95% of the Texans like the music of Prince.

Homework

Question 2

Edward is trying to prove to Bella that vampires exist. Bella thinks there is a 0.05 probability that vampires exist. She also believes that the probability that someone can sparkle like a diamond if vampires exist is 0.7, and the probability that someone can sparkle like a diamond if vampires don't exist is 0.03. Edward then goes into a meadow and shows Bella that he can sparkle like a diamond. Given that Edward sparkled like a diamond, what is the probability that vampires exist?

Homework

Question 3

The probability that Sandra will like a restaurant is 0.7. Among the restaurants that she likes, 20% have five stars on Yelp, 50% have four stars, and 30% have fewer than four stars. What other information do we need if we want to find the posterior probability that Sandra likes a restaurant given that it has fewer than four stars on Yelp?

Homework

Question 4

Robin applies for six equally competitive data science internships. He has the following prior model for his chances of getting into any given internship, π :

π	0.3	0.4	0.5	Total
$f(\pi)$	0.25	0.60	0.15	1

- (a) Let Y be the number of internship offers that Robin gets. Specify the model for the dependence of Y on π and the corresponding pmf, $f(y \mid \pi)$.
- (b) Robin got some pretty amazing news. He was offered four of the six internships! How likely would this be if $\pi = 0.3$?
- (c) Construct the posterior model of π in light of Robin's internship news.

Homework

Question 5

Whether you like it or not, cats have taken over the internet. Joining the craze, Michael has written an algorithm to detect cat images. It correctly identifies 80% of cat images as cats, but falsely identifies 50% of non-cat images as cats. Michael tests her algorithm with a new set of images, 8% of which are cats. What's the probability that an image is actually a cat if the algorithm identifies it as a cat? Answer this question by simulating data for 10,000 images.